

Geometry





About the Course

Students will discover geometric concepts by constructing drawings.

This topic is included in the following course(s): Mathematics: Grade 8

Mathematics: Grade 8

For students in elementary school, we recommend RightStart Math (Second Edition), because a) it adheres to Mason's principle that the idea always comes before the algorithm, b) it reflects the very best current research in how children learn mathematics, and c) it offers much-needed support for teachers whose own math instruction may have been lacking. Students in middle school take RightStart Geometry and also add in Pre-Algebra (Hands-On Equations) and Jacobs' Elementary Algebra.

The lessons should be done in the morning and games in the afternoon. Both mental math and games are critical parts of math instruction, so do not skip them. These lessons are paced out to help keep you on track. There are a few catch-up days built in and can be used to finish incomplete work or to review concepts. The general goal should be to complete one lesson per day. It is important that you complete one level per year, even if that means continuing during breaks, as moving slower can affect high school credits later.



Placement & Combining Tips

RightStart G Primary Skill Focus: Geometry. This course progresses through the book over two years. Continue from Grade 7.



Scheduling

GRADE	SCHEDULE INFO.	BOOKS
8	Geometry: Grade 8 2 times/week 40 min	RightStart Mathematics Level G 2nd Ed. Book Bundle

Sample Weekly View

Day 1	Day 2	Day 3	Day 4	Day 5
Mathematics: Grade 8				
Algebra 1: Part 1 Math Games: Grades 1-8	Geometry: Grade 8 Math Games: Grades 1-8	Algebra 1: Part 1 Math Games: Grades 1-8	Geometry: Grade 8	Algebra 1: Part 1



Planning & Prep

Permission to print for non-commercial use. See Alveary group use policy to use lessons in a group context.

LINKS: Click text or scan the QR code in the top corner of the lesson plan pages to view online resources associated with the lessons.

Responsibility for previewing all links rests with the teacher. All links were checked at the time of publication; however, websites change frequently and may contain objectionable content. Please report broken links by contacting us through our website.

Mathematics: Grade 8

☐ If you downloaded the RightStart ebook, print out enough copies of each page and put them in a binder for easy access.

☐ Organize math manipulatives for easy access.



Books & Resources

For book rationales and purchase options, click the Book List link or scan the QR code below.

∞ [View Book List Details](#)

Geometry: Grade 8



RightStart Mathematics Level G 2nd Ed. Book Bundle



Supplies

For supply list details and basic supplies helpful to have on hand, click the links or scan the QR code below.

∞ [View Basic Supplies](#)

(No Subject Supplies Assigned)



Quick Links

Related Course Quick Links

∞ [Extra Helpings](#)

∞ [Foundations \(See Section 10: Math\)](#)

Click THIS text
or scan the QR
code for links.



Geometry: Grade 8

How To Teach



Read

The RightStart Math Level G materials are written to the student and may be done independently. However, you may still find it very helpful to work through the lessons to familiarize yourself with the concepts.



Prepare

For Pre-Algebra, it is highly recommended that students be guided by a teacher during these lessons. Teachers may watch the video lessons to learn the concepts and how to teach them to students. If you choose to have your student work independently using the video lessons, you may still find it very helpful to work through the lessons yourself to familiarize yourself with the concepts. Teachers must stay apprised of how students are progressing and be ready to step in and help, since problems can compound quickly. If you are working through the lessons ahead of your student, note the places where you have to stop and think, because those are probably going to be places where students are going to be challenged, as well.



Growth Mindset

Resist the temptation to fall back on just teaching a memorizable procedure when the student struggles with a concept. Growth happens in the struggle. Work on resilience by modeling a growth mindset.



Leave & Return

When your student hits a wall during a lesson, stop and move on to the next subject in your day or take a break. Explain that even when you stop working on a problem, your brain continues to work on it while you do other things, and you don't even know it. Many times, when you come back to the problem the next day, it comes more easily.



Connect

Look for ways to bring math into everyday life and conversation "by the way" - through cooking, gardening, house projects, health, games, etc. Students should discover how math is useful and beautiful, and how it helps us explain and navigate the world.

Geometry: Grade 8

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Term 1

WEEK 1 40m Geometry: Grade 8 - Lesson 1

Inscribed Circles

Materials: RightStart Math Level G

PREP: Read the lesson. Work through it to acquaint yourself with the concepts. Lay out all materials.

→ PRACTICE

Lesson 89

WEEK 1 40m Geometry: Grade 8 - Lesson 2

Perpendicular Bisectors

Materials: RightStart Math Level G

PREP: Read the lesson. Work through it to acquaint yourself with the concepts. Lay out all materials.

→ PRACTICE

Lesson 90

WEEK 2 40m Geometry: Grade 8 - Lesson 3

The Amazing Nine-Point Circle

Materials: RightStart Math Level G

PREP: Read the lesson. Work through it to acquaint yourself with the concepts. Lay out all materials.

→ PRACTICE

Lesson 91

WEEK 2 40m Geometry: Grade 8 - Lesson 4

Drawing Arcs and Tangents

Materials: RightStart Math Level G

PREP: Read the lesson. Work through it to acquaint yourself with the concepts. Lay out all materials.

→ PRACTICE

Lesson 92

WEEK 3 40m Geometry: Grade 8 - Lesson 5

More Drawing Arcs

Materials: RightStart Math Level G

PREP: Read the lesson. Work through it to acquaint yourself with the concepts. Lay out all materials.

→ PRACTICE

Lesson 93

WEEK 3 40m Geometry: Grade 8 - Lesson 6

Catch-Up Day

Materials: RightStart Math Level G

→ CATCH UP

Use this time to catch up on lessons in this or another course, or explore

Geometry: Grade 8

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Term 1

Extra Helpings.

WEEK 4 40m Geometry: Grade 8 - Lesson 7

Angles 'n Arcs

Materials: RightStart Math Level G

PREP: Read the lesson. Work through it to acquaint yourself with the concepts. Lay out all materials.

→ PRACTICE

Lesson 94

WEEK 4 40m Geometry: Grade 8 - Lesson 8

Arc Length

Materials: RightStart Math Level G

PREP: Read the lesson. Work through it to acquaint yourself with the concepts. Lay out all materials.

→ PRACTICE

Lesson 95

WEEK 5 40m Geometry: Grade 8 - Lesson 9

Review and Games 9

Materials: RightStart Math Level G

PREP: Read the lesson. Work through it to acquaint yourself with the concepts. Lay out all materials.

→ PRACTICE

Lesson 96

WEEK 5 40m Geometry: Grade 8 - Lesson 10

Review and Games 9 (cont.)

Materials: RightStart Math Level G

PREP: Read the lesson. Work through it to acquaint yourself with the concepts. Lay out all materials.

→ PRACTICE

Lesson 96

WEEK 6 40m Geometry: Grade 8 - Lesson 11

Area of a Circle

Materials: RightStart Math Level G

PREP: Read the lesson. Work through it to acquaint yourself with the concepts. Lay out all materials.

→ PRACTICE

Lesson 97

Geometry: Grade 8

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Term 1

WEEK 6 40m Geometry: Grade 8 - Lesson 12

Catch-Up Day

Materials: RightStart Math Level G

→ CATCH UP

Use this time to catch up on lessons in this or another course, or explore Extra Helpings.

WEEK 7 40m Geometry: Grade 8 - Lesson 13

Finding the Area of a Circle

Materials: RightStart Math Level G

PREP: Read the lesson. Work through it to acquaint yourself with the concepts. Lay out all materials.

→ PRACTICE

Lesson 98

WEEK 7 40m Geometry: Grade 8 - Lesson 14

Babylonian Circles

Materials: RightStart Math Level G

PREP: Read the lesson. Work through it to acquaint yourself with the concepts. Lay out all materials.

→ PRACTICE

Lesson 99

WEEK 8 40m Geometry: Grade 8 - Lesson 15

Finding More Areas

Materials: RightStart Math Level G

PREP: Read the lesson. Work through it to acquaint yourself with the concepts. Lay out all materials.

→ PRACTICE

Lesson 100

WEEK 8 40m Geometry: Grade 8 - Lesson 16

Doubling Perimeter, Circumference, & Area

Materials: RightStart Math Level G

PREP: Read the lesson. Work through it to acquaint yourself with the concepts. Lay out all materials.

→ PRACTICE

Lesson 101

WEEK 9 40m Geometry: Grade 8 - Lesson 17

Pizza Problems

Materials: RightStart Math Level G

PREP: Read the lesson. Work through it to acquaint yourself with the concepts. Lay out all materials.

→ PRACTICE

Geometry: Grade 8

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Term 1	
Lesson 102	
→ NOTE Skip Lesson 103. (Save for exam.)	
WEEK 9 40m Geometry: Grade 8 - Lesson 18 <i>Catch-Up Day</i> 📄 Materials: RightStart Math Level G → CATCH UP Use this time to catch up on lessons in this or another course, or explore Extra Helpings.	
WEEK 10 40m Geometry: Grade 8 - Lesson 19 <i>Revisiting Tangrams</i> 📄 Materials: RightStart Math Level G PREP: Read the lesson. Work through it to acquaint yourself with the concepts. Lay out all materials. → PRACTICE Lesson 104	
WEEK 10 40m Geometry: Grade 8 - Lesson 20 <i>Aligning Objects</i> 📄 Materials: RightStart Math Level G PREP: Read the lesson. Work through it to acquaint yourself with the concepts. Lay out all materials. → PRACTICE Lesson 105	
WEEK 11 40m Geometry: Grade 8 - Lesson 21 <i>Catch-Up Day</i> 📄 Materials: RightStart Math Level G → CATCH UP Use this time to catch up on lessons in this or another course, or explore Extra Helpings.	
WEEK 11 40m Geometry: Grade 8 - Lesson 22 <i>Catch-Up Day</i> 📄 Materials: RightStart Math Level G → CATCH UP Use this time to catch up on lessons in this or another course, or explore Extra Helpings.	
WEEK 12 40m Geometry: Grade 8 - Lesson 23 <i>Exam Week</i> 📄 Materials: RightStart Math Level G → EXAMS Answer question(s) related to course.	

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Term 1

Questions will come from: RightStart Math Level G	
WEEK 12 40m Geometry: Grade 8 - Lesson 24 <i>Exam Week</i> Materials: RightStart Math Level G → EXAMS Answer question(s) related to course. Questions will come from: RightStart Math Level G	

Geometry: Grade 8

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Term 2

WEEK 1 40m Geometry: Grade 8 - Lesson 25

Reflecting

Materials: RightStart Math Level G

PREP: Read the lesson. Work through it to acquaint yourself with the concepts. Lay out all materials.

→ PRACTICE

Lesson 106

WEEK 1 40m Geometry: Grade 8 - Lesson 26

Rotating

Materials: RightStart Math Level G

PREP: Read the lesson. Work through it to acquaint yourself with the concepts. Lay out all materials.

→ PRACTICE

Lesson 107

WEEK 2 40m Geometry: Grade 8 - Lesson 27

Making Wheel Designs

Materials: RightStart Math Level G

PREP: Read the lesson. Work through it to acquaint yourself with the concepts. Lay out all materials.

→ PRACTICE

Lesson 108

WEEK 2 40m Geometry: Grade 8 - Lesson 28

Identifying Reflections and Rotations

Materials: RightStart Math Level G

PREP: Read the lesson. Work through it to acquaint yourself with the concepts. Lay out all materials.

→ PRACTICE

Lesson 109

WEEK 3 40m Geometry: Grade 8 - Lesson 29

Review and Games 11

Materials: RightStart Math Level G

PREP: Read the lesson. Work through it to acquaint yourself with the concepts. Lay out all materials.

→ PRACTICE

Lesson 110

WEEK 3 40m Geometry: Grade 8 - Lesson 30

Review and Games 11 (cont.)

Materials: RightStart Math Level G

PREP: Read the lesson. Work through it to acquaint yourself with the concepts. Lay out all materials.

Geometry: Grade 8

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Term 2

	→ PRACTICE Lesson 110	
WEEK 4	40m Geometry: Grade 8 - Lesson 31 <i>Translations</i> Materials: RightStart Math Level G PREP: Read the lesson. Work through it to acquaint yourself with the concepts. Lay out all materials. → PRACTICE Lesson 111	
WEEK 4	40m Geometry: Grade 8 - Lesson 32 <i>Catch-Up Day</i> Materials: RightStart Math Level G → CATCH UP Use this time to catch up on lessons in this or another course, or explore Extra Helpings.	
WEEK 5	40m Geometry: Grade 8 - Lesson 33 <i>Transformations</i> Materials: RightStart Math Level G PREP: Read the lesson. Work through it to acquaint yourself with the concepts. Lay out all materials. → PRACTICE Lesson 112	
WEEK 5	40m Geometry: Grade 8 - Lesson 34 <i>More Transformations</i> Materials: RightStart Math Level G PREP: Read the lesson. Work through it to acquaint yourself with the concepts. Lay out all materials. → PRACTICE Lesson 113	
WEEK 6	40m Geometry: Grade 8 - Lesson 35 <i>Double Reflections</i> Materials: RightStart Math Level G PREP: Read the lesson. Work through it to acquaint yourself with the concepts. Lay out all materials. → PRACTICE Lesson 114	

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Term 2

WEEK 6 40m Geometry: Grade 8 - Lesson 36

Finding the Line of Reflection

Materials: RightStart Math Level G

PREP: Read the lesson. Work through it to acquaint yourself with the concepts. Lay out all materials.

→ PRACTICE

Lesson 115

→ NOTE

Skip Lesson 116. (Save for exam.)

WEEK 7 40m Geometry: Grade 8 - Lesson 37

Finding the Center of a Rotation

Materials: RightStart Math Level G

PREP: Read the lesson. Work through it to acquaint yourself with the concepts. Lay out all materials.

→ PRACTICE

Lesson 117

WEEK 7 40m Geometry: Grade 8 - Lesson 38

Catch-Up Day

Materials: RightStart Math Level G

→ CATCH UP

Use this time to catch up on lessons in this or another course, or explore Extra Helpings.

WEEK 8 40m Geometry: Grade 8 - Lesson 39

More Centers of Rotation

Materials: RightStart Math Level G

PREP: Read the lesson. Work through it to acquaint yourself with the concepts. Lay out all materials.

→ PRACTICE

Lesson 118

WEEK 8 40m Geometry: Grade 8 - Lesson 40

More Double Reflections

Materials: RightStart Math Level G

PREP: Read the lesson. Work through it to acquaint yourself with the concepts. Lay out all materials.

→ PRACTICE

Lesson 119

Geometry: Grade 8

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Term 2

WEEK 9 40m Geometry: Grade 8 - Lesson 41

Angles of Incidence and Reflection

Materials: RightStart Math Level G

PREP: Read the lesson. Work through it to acquaint yourself with the concepts. Lay out all materials.

→ PRACTICE

Lesson 120

WEEK 9 40m Geometry: Grade 8 - Lesson 42

Lines of Symmetry

Materials: RightStart Math Level G

PREP: Read the lesson. Work through it to acquaint yourself with the concepts. Lay out all materials.

→ PRACTICE

Lesson 121

WEEK 10 40m Geometry: Grade 8 - Lesson 43

Rotational Symmetry

Materials: RightStart Math Level G

PREP: Read the lesson. Work through it to acquaint yourself with the concepts. Lay out all materials.

→ PRACTICE

Lesson 122

WEEK 10 40m Geometry: Grade 8 - Lesson 44

Catch-Up Day

Materials: RightStart Math Level G

→ CATCH UP

Use this time to catch up on lessons in this or another course, or explore Extra Helpings.

WEEK 11 40m Geometry: Grade 8 - Lesson 45

Catch-Up Day

Materials: RightStart Math Level G

→ CATCH UP

Use this time to catch up on lessons in this or another course, or explore Extra Helpings.

WEEK 11 40m Geometry: Grade 8 - Lesson 46

Catch-Up Day

Materials: RightStart Math Level G

→ CATCH UP

Use this time to catch up on lessons in this or another course, or explore Extra Helpings.

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Term 2

WEEK 12 40m Geometry: Grade 8 - Lesson 47 Exam Week Materials: RightStart Math Level G → EXAMS Answer question(s) related to course. Questions will come from: RightStart Math Level G	
WEEK 12 40m Geometry: Grade 8 - Lesson 48 Exam Week Materials: RightStart Math Level G → EXAMS Answer question(s) related to course. Questions will come from: RightStart Math Level G	

Geometry: Grade 8

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Term 3

WEEK 1 40m Geometry: Grade 8 - Lesson 49

Symmetry Connections

Materials: RightStart Math Level G

PREP: Read the lesson. Work through it to acquaint yourself with the concepts. Lay out all materials.

→ **PRACTICE**
Lesson 123

WEEK 1 40m Geometry: Grade 8 - Lesson 50

Frieze Patterns

Materials: RightStart Math Level G

PREP: Read the lesson. Work through it to acquaint yourself with the concepts. Lay out all materials.

→ **PRACTICE**
Lesson 124

WEEK 2 40m Geometry: Grade 8 - Lesson 51

Review and Games 13

Materials: RightStart Math Level G

PREP: Read the lesson. Work through it to acquaint yourself with the concepts. Lay out all materials.

→ **PRACTICE**
Lesson 125

WEEK 2 40m Geometry: Grade 8 - Lesson 52

Review and Games 13 (cont.)

Materials: RightStart Math Level G

PREP: Read the lesson. Work through it to acquaint yourself with the concepts. Lay out all materials.

→ **PRACTICE**
Lesson 125

WEEK 3 40m Geometry: Grade 8 - Lesson 53

Catch-Up Day

Materials: RightStart Math Level G

→ **CATCH UP**
Use this time to catch up on lessons in this or another course, or explore Extra Helpings.

WEEK 3 40m Geometry: Grade 8 - Lesson 54

Review for Assessment 3

Materials: RightStart Math Level G

PREP: Read the lesson. Work through it to acquaint yourself with the concepts. Lay out all materials.

Geometry: Grade 8

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Term 3

	→ PRACTICE Lesson 126	
WEEK 4	40m Geometry: Grade 8 - Lesson 55 <i>Review for Assessment 3 (cont.)</i> Materials: RightStart Math Level G PREP: Read the lesson. Work through it to acquaint yourself with the concepts. Lay out all materials. → PRACTICE Lesson 126	
WEEK 4	40m Geometry: Grade 8 - Lesson 56 <i>Review for Assessment 3 (cont.)</i> Materials: RightStart Math Level G PREP: Read the lesson. Work through it to acquaint yourself with the concepts. Lay out all materials. → PRACTICE Lesson 126	
WEEK 5	40m Geometry: Grade 8 - Lesson 57 <i>Review for Assessment 3 (cont.)</i> Materials: RightStart Math Level G PREP: Read the lesson. Work through it to acquaint yourself with the concepts. Lay out all materials. → PRACTICE Lesson 126	
WEEK 5	40m Geometry: Grade 8 - Lesson 58 <i>Review for Assessment 3 (cont.)</i> Materials: RightStart Math Level G PREP: Read the lesson. Work through it to acquaint yourself with the concepts. Lay out all materials. → PRACTICE Lesson 126	
WEEK 6	40m Geometry: Grade 8 - Lesson 59 <i>Assessment 3</i> Materials: RightStart Math Level G PREP: Read the lesson. Work through it to acquaint yourself with the concepts. Lay out all materials. → PRACTICE Lesson 127	

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Term 3

WEEK 6 40m Geometry: Grade 8 - Lesson 60 Assessment 3 (cont.) Materials: RightStart Math Level G PREP: Read the lesson. Work through it to acquaint yourself with the concepts. Lay out all materials. → PRACTICE Lesson 127	
WEEK 7 40m Geometry: Grade 8 - Lesson 61 Assessment 3 (cont.) Materials: RightStart Math Level G PREP: Read the lesson. Work through it to acquaint yourself with the concepts. Lay out all materials. → PRACTICE Lesson 127	
WEEK 7 40m Geometry: Grade 8 - Lesson 62 Assessment 3 (cont.) Materials: RightStart Math Level G PREP: Read the lesson. Work through it to acquaint yourself with the concepts. Lay out all materials. → PRACTICE Lesson 127	
WEEK 8 40m Geometry: Grade 8 - Lesson 63 Assessment 3 (cont.) Materials: RightStart Math Level G PREP: Read the lesson. Work through it to acquaint yourself with the concepts. Lay out all materials. → PRACTICE Lesson 127	
WEEK 8 40m Geometry: Grade 8 - Lesson 64 Catch-Up Day Materials: RightStart Math Level G →CATCH UP Use this time to catch up on lessons in this or another course, or explore Extra Helpings.	
WEEK 9 40m Geometry: Grade 8 - Lesson 65 Enrichment: Year-End Original Designs Materials: RightStart Math Level G PREP: Read the lesson. Work through it to acquaint yourself with the concepts. Lay out all materials.	

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Term 3

→ PRACTICE Lesson 128	
WEEK 9 40m Geometry: Grade 8 - Lesson 66 <i>Enrichment: Design Plans</i> Materials: RightStart Math Level G PREP: Read the lesson. Work through it to acquaint yourself with the concepts. Lay out all materials. → PRACTICE Lesson 129	
WEEK 10 40m Geometry: Grade 8 - Lesson 67 <i>Enrichment: Design Creation</i> Materials: RightStart Math Level G PREP: Read the lesson. Work through it to acquaint yourself with the concepts. Lay out all materials. → PRACTICE Lesson 130	
WEEK 10 40m Geometry: Grade 8 - Lesson 68 <i>Catch-Up Day</i> Materials: RightStart Math Level G → CATCH UP Use this time to catch up on lessons in this or another course, or explore Extra Helpings.	
WEEK 11 40m Geometry: Grade 8 - Lesson 69 <i>Catch-Up Day</i> Materials: RightStart Math Level G → CATCH UP Use this time to catch up on lessons in this or another course, or explore Extra Helpings.	
WEEK 11 40m Geometry: Grade 8 - Lesson 70 <i>Catch-Up Day</i> Materials: RightStart Math Level G → CATCH UP Use this time to catch up on lessons in this or another course, or explore Extra Helpings.	
WEEK 12 40m Geometry: Grade 8 - Lesson 71 <i>Exam Week</i> Materials: RightStart Math Level G → EXAMS Answer question(s) related to course.	

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Term 3

Questions will come from: RightStart Math Level G	
WEEK 12 40m Geometry: Grade 8 - Lesson 72 <i>Exam Week</i> Materials: RightStart Math Level G → EXAMS Answer question(s) related to course. Questions will come from: RightStart Math Level G	

Geometry: Grade 8

Examination

- Term 1** · RightStart Math Level G: Work on Review and Games 10 (Lesson 103).
- Term 2** · RightStart Math Level G: Work on Review and Games 12 (Lesson 116).
- Term 3** · RightStart Math Level G: Work on Assessment 3 (Lesson 127) if not finished. If finished, work on questions from another subject or play a math game.